

PROJECT REPORT

STUDENT ID MANAGEMENT SYSTEM

A Basic Python Console-Based Application

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REGISTRATION NO.	26MIB10002
PROJECT TITLE	Student ID Management System
PROGRAMMING LANGUAGE	Python
PROJECT TYPE	Console-Based Application

Submitted as a Basic Python Programming Project

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TABLE OF CONTENTS

1	Introduction
2	Objectives
3	Technologies and Requirements
4	System Features / Modules
5	Data Structure Used
6	Working of the System
7	Grade Calculation
8	Algorithm
9	Sample Output
10	Advantages
11	Limitations
12	Future Scope
13	Conclusion
14	Project Details
Appendix	Screenshots of Program Execution

1. Introduction

The Student ID Management System is a simple console-based Python application developed to store and manage student information. It allows the user to add, search, update and delete student records.

The system also provides features for student ID cards, attendance, marks, grades, fee status and a complete student report.

The project demonstrates fundamental Python concepts such as variables, lists, dictionaries, functions, loops, conditional statements, user input and basic data validation.

2. Objectives

The main objectives of this project are:

- To maintain basic student records in a structured form.
- To provide a simple menu-driven student management system.
- To demonstrate the use of Python lists and dictionaries.
- To perform CRUD operations: Create, Read, Update and Delete.
- To manage attendance, marks and fee status.
- To calculate and display grades according to marks.
- To generate a simple student ID card and student report.

3. Technologies and Requirements

Software Requirements

- Python 3.x
- Any Python editor/IDE such as IDLE, VS Code or PyCharm
- Windows/Linux/macOS operating system

Hardware Requirements

- Basic computer or laptop
- Keyboard for entering student information
- No internet connection is required to run the program

4. System Features / Modules

Module	Description
Add Student	Adds a new student and prevents duplicate Student IDs.
Show All Students	Displays the basic details of all stored students.
Search Student	Searches for a student using Student ID.
Update Student	Updates personal and academic details.
Delete Student	Deletes a student record using Student ID.
Student ID Card	Displays a formatted ID card for a selected student.
Attendance	Updates attendance percentage between 0 and 100.
Marks	Stores marks percentage between 0 and 100.
Grade	Calculates grade from the student's marks.
Fee Status	Changes fee status to Paid or Not Paid.

Module	Description
Student Report	Displays attendance, marks, grade and fee status together.

5. Data Structure Used

The program uses a list named **students**. Each student is stored as a Python dictionary. This makes it easy to access individual details using keys such as id, name, course, branch, attendance, marks and fees.

```
student = {
    "id": sid,
    "name": name,
    "age": age,
    "gender": gender,
    "course": course,
    "branch": branch,
    "year": year,
    "phone": phone,
    "email": email,
    "attendance": 0,
    "marks": 0,
    "fees": "Not Paid"
}
```

6. Working of the System

When the program starts, it displays a menu containing twelve options. The user selects an option by entering its number. The corresponding function is then executed. After the function finishes, the menu is displayed again until the user selects Exit.

Basic working flow:

1. Start the program.
2. Display the Student ID Management menu.
3. Accept the user's choice.
4. Execute the selected function.
5. Display the result or an appropriate error message.
6. Return to the main menu.
7. Stop the program when the user selects option 12.

7. Grade Calculation

Marks	Grade
90 – 100	A+
80 – 89	A
70 – 79	B
60 – 69	C
50 – 59	D
40 – 49	E
Below 40	F

8. Algorithm

Algorithm for the main program:

- Step 1: Initialize an empty list called students.
- Step 2: Display the main menu.
- Step 3: Read the user's choice.
- Step 4: Call the function corresponding to the selected option.
- Step 5: Perform the requested operation on the student list.
- Step 6: Display the result.
- Step 7: Repeat Steps 2–6 until the Exit option is selected.
- Step 8: End the program.

9. Sample Output

Main Menu

```
=====
STUDENT ID MANAGEMENT
=====
```

1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
12. Exit

Example Student Report

Student ID: 26MIB10002

Name: Aryan Gupta

Course: B.Tech

Branch: CSE

Year: 1

Attendance: 85 %

Marks: 82 %

Grade: A

Fees: Paid

10. Advantages

- Simple and easy-to-use menu-driven interface.
- Uses basic Python concepts suitable for beginners.
- Provides several student management operations in one program.
- Includes basic validation for duplicate IDs, attendance and marks.
- Does not require any external database or library.

11. Limitations

- Student data is stored only in memory.
- All records are lost when the program is closed.
- It is a console-based application and has no graphical user interface.
- It does not use a database for permanent storage.
- Input validation can be expanded for phone numbers, emails and other fields.

12. Future Scope

- File handling or a database such as MySQL/SQLite for permanent storage.
- A graphical user interface using Tkinter.
- Login and administrator authentication.

- Student photo support and printable ID cards.
- Subject-wise marks and attendance.
- Automatic report generation.
- Search by name, course or branch.

13. Conclusion

The Student ID Management System successfully demonstrates how Python can be used to build a basic student record management application. The project combines lists, dictionaries, functions, loops and conditional statements to perform useful operations such as adding students, searching records, updating information, calculating grades, managing fees and generating student reports. It provides a strong foundation for developing a more advanced student management system using files, databases and graphical interfaces.

14. Project Details

Student Name	Aryan Gupta
Registration Number	26MIB10002
Project	Student ID Management System
Language	Python

APPENDIX — PROGRAM EXECUTION SCREENSHOTS

The following screenshots show the actual execution of the Student ID Management System, including student creation, listing, search, update, ID card, attendance, marks, grade, fee status and final student report.

Screenshot 1

```
=====
                        STUDENT ID MANAGEMENT
=====
1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
12. Exit
```

Enter your choice: 1

```
--- Add Student ---
Enter Student ID: 10006
Enter Name: Raunak
Enter Age: 18
Enter Gender: Male
Enter Course: Int Mtech
Enter Branch: Ai bioinformatics
Enter Year: 1st
Enter Phone Number: 9098712
Enter Email: raunak1234
Student added successfully!
```

```
=====
                        STUDENT ID MANAGEMENT
=====
1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student
```

Screenshot 2

```
=====
                        STUDENT ID MANAGEMENT
=====
1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
12. Exit

Enter your choice: 2

--- Student List ---

Student ID: 10006
Name: Raunak
Course: Int Mtech
Branch: Ai bioinformatics
Year: 1st
Phone: 9098712
-----

=====
                        STUDENT ID MANAGEMENT
=====
1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
12. Exit

Enter your choice: 3

--- Search Student ---
```

Screenshot 3

```
=====
1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
12. Exit

Enter your choice: 3

--- Search Student ---
Enter Student ID: 10006

Student Found!
ID: 10006
Name: Raunak
Age: 18
Gender: Male
Course: Int Mtech
Branch: Ai bioinformatics
Year: 1st
Phone: 9098712
Email: raunak1234
Attendance: 0 %
Marks: 0 %
Fees: Not Paid

=====
                        STUDENT ID MANAGEMENT
=====
1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
```

Screenshot 4

```
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
12. Exit

Enter your choice: 4

--- Update Student ---
Enter Student ID: 10002
Student not found.

=====
                STUDENT ID MANAGEMENT
=====

1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
12. Exit

Enter your choice: 4

--- Update Student ---
Enter Student ID: 10006
Enter the new details.
Name: Aryan Gupta
Age: 18
Gender: Male
Course: Integrated Mtech
Branch: Ai bioinformatics
Year: 1
Phone: 987612
Email: aryannnn1317
Student details updated.

=====
```

Screenshot 5

```
Branch: Ai bioinformatics
Year: 1
Phone: 987612
Email: aryannnn1317
Student details updated.
```

```
=====
          STUDENT ID MANAGEMENT
=====
```

1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
12. Exit

Enter your choice: 6

```
--- Student ID Card ---
Enter Student ID: 10006
```

```
=====
          COLLEGE ID CARD
=====
```

```
Student ID : 10006
Name       : Aryan Gupta
Course     : Integrated Mtech
Branch     : Ai bioinformatics
Year       : 1
Phone      : 987612
=====
```

```
=====
          STUDENT ID MANAGEMENT
=====
```

1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student

Screenshot 6

4. Update Student
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
12. Exit

Enter your choice: 7

--- Update Attendance ---
Enter Student ID: 10006
Enter attendance percentage: 100
Attendance updated.

```
=====
      STUDENT ID MANAGEMENT
=====
1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
12. Exit
```

Enter your choice: 8

--- Enter Marks ---
Enter Student ID: 10006
Enter marks percentage: 98
Marks added successfully.

```
=====
      STUDENT ID MANAGEMENT
=====
1. Add Student
2. Show All Students
3. Search Student
4. Update Student
```

Screenshot 7

```
Enter your choice: 9

--- Student Grade ---
Enter Student ID: 10006
Name: Aryan Gupta
Marks: 98.0 %
Grade: A+

=====
                STUDENT ID MANAGEMENT
=====

1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
12. Exit

Enter your choice: 10

--- Fee Status ---
Enter Student ID: 10006
1. Paid
2. Not Paid
Enter choice: 1
Fee status updated.

=====
                STUDENT ID MANAGEMENT
=====

1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
```

Screenshot 8

```
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
12. Exit

Enter your choice: 11

--- Student Report ---
Enter Student ID: 10006

Student Report
-----
Student ID: 10006
Name: Aryan Gupta
Course: Integrated Mtech
Branch: Ai bioinformatics
Year: 1
Attendance: 100.0 %
Marks: 98.0 %
Grade: A+
Fees: Paid
-----

=====
          STUDENT ID MANAGEMENT
=====

1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Student ID Card
7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
12. Exit

Enter your choice: 12

Thank you for using the Student ID Management System!
```